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Motivational effects on the processing of delayed intentions in the anterior prefrontal cortex

Jovita Bruening[a,b,d], Vera U. Ludwig[a,c,d], Lena M. Paschke[a,c,d], Henrik Walter[a,c,e,1], Christine Stelzel[a,c,d,e,f,*,1]

[a] Department of Psychiatry and Psychotherapy, Charité Universitaetsmedizin Berlin, Charitéplatz 1, 10117, Berlin, Germany
[b] Work, Engineering and Organizational Psychology Technische Universitaet Berlin, Marchstr. 12, F7, 10587 Berlin, Germany
[c] Berlin School of Mind and Brain, Humboldt Universitaet zu Berlin, Luisenstr. 56, 10117 Berlin, Germany
[d] Department of Psychology, Humboldt Universitaet zu Berlin, Rudower Chaussee 18, 12489 Berlin, Germany
[e] Berlin Center for Advanced Neuroimaging, Charité Universitaetsmedizin Berlin, Charitéplatz 1, 10117 Berlin, Germany
[f] International Psychoanalytic University, Stromstr. 1, 10555 Berlin, Germany

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ABSTRACT

Delaying intentions bears the risk of interference from distracting activities during the delay interval. Motivation can increase intention retrieval success but little is known about the underlying brain mechanisms. Here, we investigated whether motivational incentives (monetary reward) modulate the processing of delayed intentions in the anterior prefrontal cortex (aPFC), known to be crucial for intention processing. Using a mixed blocked and event-related functional Magnetic Resonance Imaging design, we specifically tested whether reward affects intention processing in the aPFC in a transient or in a sustained manner and whether this is related to individual differences in retrieval success.

We found a generalized effect of reward on both correct intention retrieval and ongoing task performance. Fronto-parietal regions including bilateral lateral aPFC showed sustained activity increases in rewarded compared to non-rewarded blocks as well as transient reward-related activity during the storage phase. Additionally, individual differences in reward-related performance benefits were related to the degree of transient signal increases in right lateral aPFC, specifically during intention encoding.

This suggests that the ability to integrate motivational relevance into the encoding of future intentions is crucial for successful intention retrieval in addition to general increases in processing effort. Bilateral aPFC is central to these motivation-cognition interactions.

Introduction

A common demand we have to meet in our everyday life is to hold a formed intention in mind, which cannot be performed immediately due to ongoing activities. At a future point in time, these stored intentions need to be remembered, when the circumstances permit or require to complete the intended action. This kind of memory for activities to be executed in the future (delayed intentions) is also referred to as prospective memory (Brandimonte et al., 1996; Meacham and Dumitru, 1976). Importantly, the retrieval of intentions and hence the achievement of the corresponding goals decisively relies on the degree to which an individual is motivated as has been shown in behavioral prospective memory studies (Brandimonte et al., 2010; Cook et al., 2015; Einstein et al., 2005; Kliegel et al., 2004; Meacham and Singer, 1977; Walter and Meier, 2014). Consequently, combining intentions with motivational significance enables individuals to create and realize plans based on their priorities and thus is essential for shaping the not so distant future (Penningroth and Scott, 2007).

The behavioral effects of motivation on intention retrieval increasingly attract attention (Penningroth and Scott, 2013; Walter and Meier, 2014), but little is known about the underlying brain mechanisms (but see Rea et al., 2011). Two major streams of research investigated the neurocognitive implementation of intention processing in general. On the one hand, research on volitional selection of intentions and the representation of their content indicates a crucial role of the dorsomedial prefrontal cortex for the processing of self-generated intentions (Brass

* Corresponding author. International Psychoanalytic University, Stromstr. 1, D-10555 Berlin, Germany.
E-mail address: christine.stelzel@ipu-berlin.de (C. Stelzel).

[1] Equal contribution.

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et al., 2013; Brass and Haggard, 2007; Haynes et al., 2007; Momennejad and Haynes, 2013). On the other hand, prospective memory research consistently showed a crucial role of the lateral anterior prefrontal cortex (aPFC; Brodmann area [BA] 10) for the processing of instructed intentions to be retrieved based on a specific event or point in time (Burgess et al., 2001, 2011; Okuda et al., 1998). This lateral aPFC involvement has been further specified as being related to strategic monitoring for intention-relevant target events (Beck et al., 2014; McDaniel et al., 2013; Reynolds et al., 2009), processes related to the storage of the intention (Gilbert, 2011) and the establishment of an intention retrieval mode (Guynn, 2003). The study by Gilbert (2011) is of particular interest for the investigation of motivational modulations, as it is one of the rare studies using an event-related functional Magnetic Resonance Imaging (fMRI) paradigm to separate different phases of processing delayed intentions: (1) encoding of the intention into memory, (2) storage of the intention during a delay with an interfering ongoing task and (3) self-initiated retrieval of the intention based on a pre-defined event. The study showed that it is particularly the storage of intentions, which is selectively related to enhanced lateral aPFC activity compared to pure ongoing task performance. The novel options given by the Gilbert paradigm inspired the design of the present study, aiming at locating motivational effects on successful intention retrieval with regard to the different processing phases.

With respect to motivation-cognition interactions, recent evidence suggests that rewards strongly affect neurocognitive processing in various domains of cognition (Charron and Koechlin, 2010; Jimura et al., 2010; Krawczyk et al., 2007; Paschke et al., 2015; Soutschek et al., 2015; Wittmann et al., 2005). The available data widely support the view that reward enhances top-down processing in cognitive tasks, thereby recruiting control-related regions in the medial and lateral prefrontal cortex (PFC) as well as further task- and stimulus-related regions (Pessoa and Engelmann, 2010). Importantly, there is evidence that reward effects can be implemented via two different processing modes (Braver, 2012; Engelmann et al., 2009; Jimura et al., 2010). In a *sustained mode,* reward-related controlled processing is enhanced independent of the concrete occurrence of stimuli requiring controlled processing throughout a task block, indicating that proactive control mechanisms might be in play. In contrast, *transient* reward-related increases related to specific processing requirements of single stimuli provide a temporary, reactive mechanism to enhance specific information processing steps depending on the prospect of a reward. Hence, control mechanisms might be recruited to a higher degree when a reward is provided depending on the actual requirements at hand. In the case of intention processing, rewards might enhance the degree of goal relatedness assigned to the respective intentions (Penningroth and Scott, 2007). This, in turn can be assumed to enhance processing in brain regions associated with intention processing. Anterior regions of the lateral PFC have been repeatedly found to show a reward-related sustained activity increase in cognitive control tasks (Braver et al., 2003; Jimura et al., 2010; Locke and Braver, 2008). With respect to transient effects, the neuroanatomical location of reward effects seems to be more dependent on the specific task demands. For example, working memory studies showed effects of reward in stimulus-related regions specifically during the encoding of the stimuli (Krawczyk et al., 2007). Also, long-term memory studies using the subsequent-memory paradigm showed the crucial role of a reward-related modulation during encoding for later retrieval (Adcock et al., 2006; Wittmann et al., 2005), indicating that hippocampal and midbrain activity during the encoding of rewarded stimuli can predict retrieval success. Furthermore, transient activity increases related to interference processing in cognitive control tasks has been shown to be modulated by reward in stimulus-related regions but also in task-related regions in the left posterior lateral frontal cortex (Padmala and Pessoa, 2011; Paschke et al., 2015; Soutschek et al., 2015).

For intention processing, it is unclear whether motivation enhances intention retrieval via sustained processing increases in task-related regions in the lateral aPFC and/or by modulating intention processing in a transient manner by selectively enhancing specific processing phases, that is, the encoding, storage or retrieval of an intention. Here, we used fMRI to address this question, focusing on the association of reward-related processing increases in task-related regions with actual behavioral improvements in retrieving intentions depending on the expectation of a monetary reward. We expected reward to be associated with a sustained increase of activity in lateral aPFC regions. Additionally, based on the outlined working memory and long-term memory research, we expected that reward-related intention retrieval success would depend on transient activity changes in task-relevant regions in the lateral aPFC during the encoding of rewarded intentions.

To investigate this, we created a new version of the paradigm used in the study by Gilbert (2011), including a motivational manipulation by providing monetary incentives for correct intention retrievals in half of the blocks in a mixed blocked and event-related fMRI design. On a behavioral level, we hypothesized that increased motivation based on rewards would increase intention retrieval performance. Concerning the fMRI data, the current design enabled us to investigate not only (i) whether intention-related regions in the aPFC are affected by motivation in a sustained manner, but also (ii) whether transient reward-related modulations in the aPFC are present at specific processing stages (encoding, storage or retrieval) and finally (iii) whether sustained and/or transient motivational effects are related to behavioral improvements.

Materials and methods

Participants

A total of 24 healthy right-handed volunteers with no history of neurological or psychiatric disorders took part in this study. The datasets of two participants were excluded from the analysis due to scanner artifacts and an incidental finding of brain abnormalities, resulting in a final sample of 22 participants (12 female, mean age = 26.8 years, standard deviation = 3.7 years, range = 22–34 years). All participants were native German speakers with normal or corrected-to-normal vision. The experiment was approved by the Ethics Committee of the Department of Psychology, Humboldt University Berlin and participants provided written informed consent before taking part according to the Declaration of Helsinki. They received 10 Euro/hour and additionally performance-dependent monetary reward.

Experimental design

A new variant of the paradigm introduced by Gilbert (2011) was developed as a mixed blocked and event-related design (Fig. 1A and B). A trial consisted of a sequence of five to eight words. Participants performed an ongoing one-back task throughout the experiment, indicating on each stimulus whether a currently shown word matched the word presented as previous stimulus by pressing a left (for one-back non-targets) or right button (for one-back targets) with the index and middle fingers of their right hand, respectively. Note that we applied a one-back task instead of the two-back task used in Gilbert's study due to high error rates (ERs) observed during a two-back version in a pilot study conducted in our laboratory. The intention task was embedded within this one-back task. While stimuli in the ongoing task were surrounded by a grey frame, a green colored frame defined a word to be encoded as the target of the delayed intention ('intention encoding'). These intention-encoding stimuli were also part of the one-back task, but always one-back non-targets, so that participants had to press the non-target button with their right index finger. The delayed intention instruction was to press a third button with the right ring finger as soon as the word originally presented in the green frame was presented again in a grey frame in the stream of words ('intention retrieval'). Accordingly, they were then required to interrupt their ongoing performance of the one-back task in order to respond to the intention target word. Note that delayed intention target words were presented in a grey frame on their second appearance just

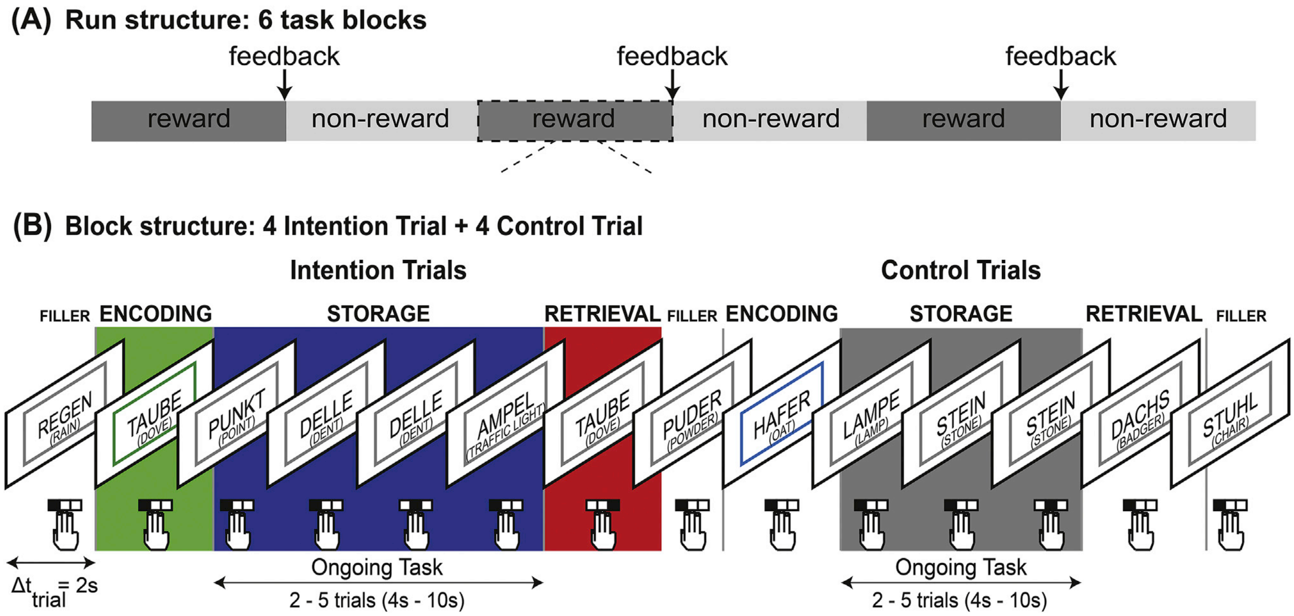

Fig. 1. Experimental paradigm, behavioral results and intention-related network. (A) Reward for correct intention retrieval (10 ct vs. no reward) was manipulated block-wise via instructions at the beginning of each block. (B) Within each block, participants performed an ongoing one-back working memory task on word identity throughout the block. Intention trials were embedded into the one-back task and alternated with control trials. Green-colored frames around the words defined intention-encoding trials. After a variable number of one-back trials, the word was presented again and participants had to retrieve the intention to respond with a different response button. In control trials the blue-colored frame was not associated with any intention.

like all of the ongoing task stimuli. Therefore, a usual one-back response would have been possible, if intention target detection failed. Thus, interrupting the ongoing task performance on the occurrence of the appropriate stimulus had to be self-initiated and was not directly cued by stimulus features.

Additionally, participants were informed that they only needed to store this intention until the next colored frame occurred. The intention target word was repeated after two to five one-back trials after the encoding. Thus, the duration of the storage phase ('intention storage') was varied pseudorandomly, providing a temporal jitter to distinguish the activity associated with the storage phase from activity associated with the encoding and retrieval phases.

In addition to these intention trials, control trials were inserted in an alternating fashion to control for salience-related effects due to the change of the color of the frame. Identically to intention trials, control trials started with a colored frame. However, the blue color associated with control trials indicated no additional task requirements. Accordingly, participants were instructed to continue with the performance of the ongoing one-back task until the next intention trial. The number of one-back trials following the colored frame was matched to the intention trials, also varying in a non-predictable way between two to five trials. Consequently, control trials were perceptually similar to intention trials, but they did not require encoding, storage or retrieval of an intention. Thus, transient processing related to the encoding, storage and retrieval of the intention should not occur during control trials as no intention-related stimuli were presented.

The motivational manipulation was implemented by alternating rewarded and non-rewarded task blocks, each containing four intention trials and four control trials in an alternating order, counterbalanced across and within participants. In rewarded blocks, additional monetary incentives were provided, depending on correct intention retrievals. Reward-related enhanced sustained activity was expected throughout the rewarded blocks, that is, independent of the occurrence of intention-related stimuli. Participants could gain 10 cents for each correct intention retrieval response. Thereby, a bonus of up to 6 € could be achieved in the whole experiment. At the beginning of each block, an instruction cue was presented informing participants whether they could gain money in the following block ("Gewinnblock", German for "gain block") or not ("Basisblock", German for "baseline block"). After eight intention/control trials, a fixation cross was shown, completing the task block. At the end of rewarded blocks, feedback was given on the amount of money earned in the last block and on the current total amount. Furthermore, a criterion was included to prevent individuals from engaging only in the intention task in order to achieve the maximal bonus: participants only received the payment for their performance in a rewarded block if their accuracy in the ongoing task reached at least 80%. Participants were informed in the feedback trial in case their ongoing task performance was below 80%.

Procedure

The whole experiment consisted of five runs each containing six blocks, that is three rewarded and three non-rewarded blocks, which were presented in alternating order. The type of the first block was counterbalanced across runs and individuals as well as the type of the trial (intention vs. control). In the first trial of each block and between every intention and control trial, that is, after the intention retrieval or according control trial, another grey-framed word was presented as a filler trial. Altogether, participants viewed 314 stimuli in each run at the center of a screen, viewable via a mirror inside the MRI scanner. All stimuli were shown for 1 s each followed by a fixation cross presented for 1 s. Individuals could respond during the presentation of the word and the following fixation cross. Instruction cues, fixation crosses (at the end of each block) and feedback were presented for 6 s each. This yielded 124 s for rewarded blocks and 118 s for non-rewarded blocks, respectively. Each run lasted approximately 13 min, including the presentation of a fixation cross (baseline period) for 20 s before and after the six blocks.

Prior to the main experiment, participants were instructed and practiced the tasks on a notebook, outside the scanner. At first, only the ongoing one-back task was introduced for better understanding. Participants were given 25 practice trials to familiarize themselves with this procedure. Subsequently, the ongoing task combined with the intention task was trained for 20 trials (i.e. 10 intention trials and 10 control trials).

Thereby, this practice version minimized later learning effects. In addition, participants performed a short experiment inside the scanner before the main experiment, including a one-back task and a zero-back task (not reported here).

Stimulus material

Words were selected from the Berlin Affective Word List (BAWL), a database containing emotional valence, emotional arousal and imageability ratings for more than 2.200 German words (Vo et al., 2006). The stimuli were chosen with the constraint that they were nouns, five letters long, characterized by neutral emotional valence (range = −1–+1 points on a scale from −3 to 3), high imageability (range = 3–6 points on a scale from 1 to 7), low emotional arousal (range = 1–3 points on a scale from 1 to 5) and high-frequency of appearance per million words (range = 0–636 total frequency of appearance per million words[2]). After the experiment, all participants rated each word in an online questionnaire, which was accessible at QuestBack Unipark. This online survey contained the same kind of ratings scales as the original BAWL, that is, a 7-point scale ranging from −3 (very negative) through 0 (neutral) to +3 (very positive) for emotional valence, a 7-point imageability scale ranging from 1 (low imageability) to 7 (high imageability) and emotional arousal was rated on a 5-point scale extending from 1 (low arousal) to 5 (high arousal). Results of our survey confirmed that the imageability of the used stimuli was high (mean [M] = 6.17), the emotional valence was neutral (M = 0.29) and the emotional arousal was low (M = 1.86), thus matching the values given in BAWL.

Selected stimuli were randomly and exclusively assigned to one of the three parts of the experiment: intention/control task (48 words), one-back task (64 words) and practice version (64 words). For the main task (314 stimuli/run), pseudorandomized lists of words were created for each run. Concerning the ongoing task, 242 stimuli were presented. These were selected from the list of 64 words: 50 one-back targets (maximal number of presentations per run: 3) and 192 one-back non-targets (maximal number of repetitions per run: 6) were presented. In addition, 24 intention-encoding stimuli, 24 control-encoding stimuli and 24 intention-retrieval stimuli were included according to the structure described above. All stimuli in intention/control encoding trials and the corresponding intention retrieval trials occurred only once in each run. Note that the intention- and control-related words used in rewarded vs. non-rewarded blocks did not differ in their properties on either of the four scales of the BAWL (all ps > .06).

The exact positions of all words in each list were pseudo-randomized as well as the order of the varying duration of the storage phases (3–6 words). The order of the words was checked for phonological similarities (e.g., rhymes) and to prevent consecutively shown words to form compound words. All words were presented in capital letters, Arial typeface.

Behavioral data analyses

Mean reaction times (RTs) and mean levels of ERs in the intention task and the ongoing one-back task were analyzed using SPSS 20 (IBM SPSS Statistics). RTs were calculated separately for correct and incorrect trials of the intention task and the one-back task. Regarding the intention task, ERs comprised all intention retrieval trials which were missed or treated as trials of the ongoing task by the participants. The ongoing task ERs included missed one-back responses as well as false responses to one-back target and non-target stimuli. Note that false ongoing task responses also comprised false alarms with regard to the intention task (i.e. ring finger response), which, however, occurred very rarely (0.03% of all responses). In addition, a more detailed descriptive analysis of the ERs by error type and task phase and RTs by task phase can be found in the Supplement (Tables S4 and S5).

Paired-samples t-tests were performed on correct RTs and ERs of intention retrieval trials comparing rewarded and non-rewarded trials. Separate two-way repeated measures analyses of variance (ANOVA) were calculated on correct RTs and ERs of one-back trials, using reward (reward vs. no-reward) and intention (intention vs. control) as within subject factors. In addition, we calculated repeated measures ANOVAs with the factors reward (reward vs. non-reward) and task performance (correct vs. incorrect) on RTs separately for the intention task and the ongoing task in order to test for a general promotion of RTs by reward. For the ongoing task, also the factor intention processing (intention trials vs. control trials) was included in this analysis. A 2 (encoding vs. storage) X 2 (reward vs. non-reward) X 2 (intention vs. control) repeated-measures ANOVA on RTs of correct non-target responses was calculated to test whether RTs of responses in the ongoing task (non-target trials) were slower when an intention was encoded concurrently.

fMRI acquisition

All images were acquired using a 3 Tesla Siemens TIM Trio MRI scanner with blood oxygen level-dependent (BOLD) contrast and a 12-channel head coil at the Berlin Center for Advanced Neuroimaging (BCAN, Berlin, Germany). Head motion was limited using foam head padding for comfortable stabilization. Participants were provided with earplugs and headphones to dampen scanner noise and enable communication. Experimental stimuli were presented with Presentation software (Neurobehavioral Systems, http://www.neurobs.com/) and projected onto a screen positioned at the head end of the bore, viewable through a mirror attached to the head coil. Behavioral performance was also recorded with Presentation software via a fiber optic response keypad consisting of four buttons. Participants were instructed to use the inner three buttons of the keypad.

T2-weighted echo-planar images (EPI) were performed in five runs (time echo TE = 25 ms, flip angle = 78°, Field of View = 24 cm, Matrix size = 64 × 64, TR = 2 s, slice thickness = 3 mm, inter-slice gap = 0.75 mm). Due to a malfunction of the projector the first run of one dataset was excluded. The main task contained 392 volumes. All functional volumes comprised 32 axial slices. All slices were oriented approximately to the anterior-posterior commissure plane and then rotated −20° to provide optimal coverage of the prefrontal cortex (Deichmann et al., 2003). Furthermore, field maps were acquired between the second and the third run of the main experiment using the same slice prescriptions as for functional scans. After the experiment, a structural T1-weighted 3-D MPRAGE scan was performed (matrix size 256 × 256 x 192, slice thickness: 1.0 mm). Anatomical images were used for the normalization of the functional data to the Montreal Neurological Institute (MNI) atlas space.

fMRI data analyses

All analyses of functional data were performed with Statistical Parametric Mapping software (SPM8, http://www.fil.ion.ucl.ac.uk/spm). The functional volumes of each participant were first realigned and unwarped, then corrected for different slice acquisition times and co-registered to the anatomical image. Subsequent preprocessing stages included segmentation and spatial normalization of the functional datasets into standard MNI space. Finally, functional data were smoothed with an 8 mm FWHM Gaussian kernel and high-pass filtered to a cut-off of 1/256 Hz during statistical analyses. An analytic design matrix was constructed modeling onsets and duration of each trial phase for each participant and a general linear model (GLM) for serially auto-correlated data was applied (Friston et al., 1995). The functional volumes acquired during the five runs of the main task were treated as separate time series.

Visual inspection for motion outliers in any dimension from one volume to the next and during every run was performed using the artifact detection toolbox (RapidArt, http://www.nitrc.org/projects/artifact_

[2] Note that some very common words of daily use like "Jacke" (German for: jacket) had a frequency of appearance of zero in BAWL due to reliance on mostly written language.

detect). Thereby, two runs of one participant were discarded due to head motion >3 mm within run, resulting in 3 runs included in the analyses for this participant.

First, the fMRI data were analyzed with an event-related design including the factors task phase (encoding, storage, retrieval), intention (intention trial vs. control trial) and reward (rewarded vs. non-rewarded) resulting in a total of twelve task-related regressors. The encoding as well as the retrieval trials were modeled as delta functions, whereas the storage periods were represented by box-car functions varying in their duration between 4s and 10s. Additional covariates were included for filler trials, one-back errors, instructions, feedback, and fixation periods between the blocks as well as for fixation periods as a baseline condition. Intention trials containing incorrect intention retrieval (including misses) were not considered in further analyses and formed separate regressors for error trials. All covariates were convolved with a canonical hemodynamic response function. These regressors, together with the six movement parameters as covariates of no interest and the mean over scans, comprised the full model for each run. Regression parameters were estimated using the classical Restricted Maximum Likelihood (ReML) algorithm. To assess intention-related processing per task phase, we defined the relevant contrasts per participant by contrasting the respective regressors per task phase in no-reward blocks between intention trials and control trials. These contrasts were then subjected to one-sample t-tests. Images were thresholded with $p < .05$, FWE-cluster-corrected ($p < .001$ at the voxel level) on a whole brain level. In addition, the combined signal of all three task phases in the non-rewarded blocks was used for the definition of a mask for the evaluation of brain-behavior correlations within frontal intention-related regions. For this, a sum image of all three phase-specific images (thresholded at $p < .001$, $k > 5$) was calculated and saved as a binary image, thus including all voxels that showed an intention effect during either encoding, storage or retrieval. Based on our a-priori interest in frontal regions, analyses with this mask were restricted to the frontal lobes via inclusive masking using the frontal lobe mask provided by xjview (http://www.alivelearn.net/xjview).

To assess transient reward-related effects, the different task phases of intention and control trials in rewarded blocks were jointly contrasted with the respective regressors in non-rewarded blocks on the single subject level. Furthermore, we tested for intention-specific transient reward effects by calculating positive interaction contrasts. Again, one-sample t-tests were used to test for group effects of these contrasts on the second level.

To assess sustained reward-related effects, these were modeled in a block design including one regressor for the rewarded blocks and another regressor for the non-rewarded blocks, modeling the duration of the whole block. Again, additional covariates were included for instructions, feedback, and fixation periods between the blocks as well as for fixation periods as the baseline condition. The block design allowed examining for sustained motivational effects by contrasting the two regressors (reward vs. no-reward) on the first level and then applying a one-sample *t*-test on the second level. To correct for multiple comparisons, a threshold of $p < .05$, FWE-cluster-corrected was used in a whole brain analysis. All structural labels provided for functional activity clusters in the supplementary tables were based on grey matter regions with a size of more than 10 voxels assigned to the activity cluster in xjview.

Brain-behavior correlations

To assess whether there were reward-related signal increases within the frontal intention network that were directly associated with reward-related behavioral improvements in intention retrieval performance, we performed a voxel-wise correlation with the individual reward-related behavioral effect (% missed intention retrievals in non-rewarded blocks minus rewarded blocks). This was done via a covariate analysis within SPM (*t*-test) to assess whether specific sub-regions of the frontal intention network were associated with behavioral improvements in a transient or sustained manner. In addition, the transient correlation analysis was also performed for the control trials to assess the specificity of the effects for intention processing within the frontal intention-related mask. Small volume correction (SVC) via the frontal intention-related mask was applied with a threshold of $p < .05$, FWE-SVC, cluster corrected.

Results

Behavioral data

Intention task

A paired-sample *t*-test on mean levels of ERs in intention retrieval trials revealed a significant effect of motivation on intention retrieval performance, $t(21) = 5.12$, $p < .001$, $d = 0.79$, indicating that participants committed fewer errors in rewarded blocks (Mean (M) = 15.92%, SE = 2.33) than in non-rewarded blocks (M = 25.20%, SE = 2.61; see Fig. 2A, left panel). Mean RTs did not differ significantly between intention retrieval trials in rewarded blocks (M = 556.97 ms, SE = 17.22) and intention retrieval trials in non-rewarded blocks (M = 562.04 ms, SE = 18.89), $t(21) = 0.76$, $p = .45$, $d = 0.06$. The ANOVA comprising the factors reward and task performance on RTs confirmed that no significant main effect of reward on RTs was present. While RTs for incorrect intention retrieval trials were generally faster than for correct trials ($p = .01$), no significant interactions of reward and task performance occurred (all $p > .13$).

Ongoing task

A 2 (reward vs. non-reward) X 2 (intention vs. control) repeated-measures ANOVA on ERs revealed a significant main effect of intention processing, $F(1, 21) = 8.64$, $p = 0.008$, $\eta^2 = 0.29$, pointing to an interference effect due to the presence of the delayed intention with higher ERs in ongoing one-back trials during intention storage (M = 5.89%, SE = 0.94) compared to pure ongoing one-back control trials (M = 4.49%, SE = 0.98; see Fig. 2A, right panel). In addition, we found a significant main effect of reward on ERs in ongoing one-back trials, $F(1,21) = 11.21$, $p = 0.003$, $\eta^2 = 0.35$, with lower ERs in rewarded blocks (M = 4.59%, SE = 0.96) compared to non-rewarded blocks (M = 5.79%; SE = 0.94). The two factors did not interact ($p = .81$, $\eta^2 = 0.003$). The corresponding ANOVA on mean RTs in correct one-back trials revealed a significant main effect of intention processing, $F(1,21) = 48.80$, $p < 0.001$, $\eta^2 = 0.7$, with slower RTs on intention trials (M = 501.57 ms, SE = 15.68) compared with control trials (M = 462.69 ms, SE = 15.80). There was no significant effect of reward ($p = .075$, $\eta^2 = 0.14$) and no interaction effect ($p = .79$, $\eta^2 = 0.003$).

The ANOVA comparing RTs for intention and control trials in rewarded and non-rewarded blocks separately for correct and incorrect one-back responses showed also only a main effect of intention processing ($p < .001$), in terms of slower responses in the ongoing task when an intention was stored, and an interaction between task performance (correct vs. incorrect) and intention processing ($p = .001$), but no main effect of or interaction with the factor reward (all $p > .31$).

A closer inspection of the ongoing task ERs per task phase, reward and intention condition, separated for the according false alarms and misses revealed ERs below 1% for all conditions except for misses (M = 2.4%, SE = 0.52) and false alarms of n-back non-targets (M = 11.16%, SE = 2.04). Accordingly, there were too few trials for further statistical analyses (but see Supplementary Table S4 for detailed descriptive data). For RTs, we compared correct non-target responses (see Supplementary Table S5) in more detail to test for interference effects between task phases. A 2 (encoding vs. storage) X 2 (reward vs. non-reward) X 2 (intention vs. control) repeated-measures ANOVA on non-target RTs showed a significant main effect of task phase, $F(1,21) = 98.79$,

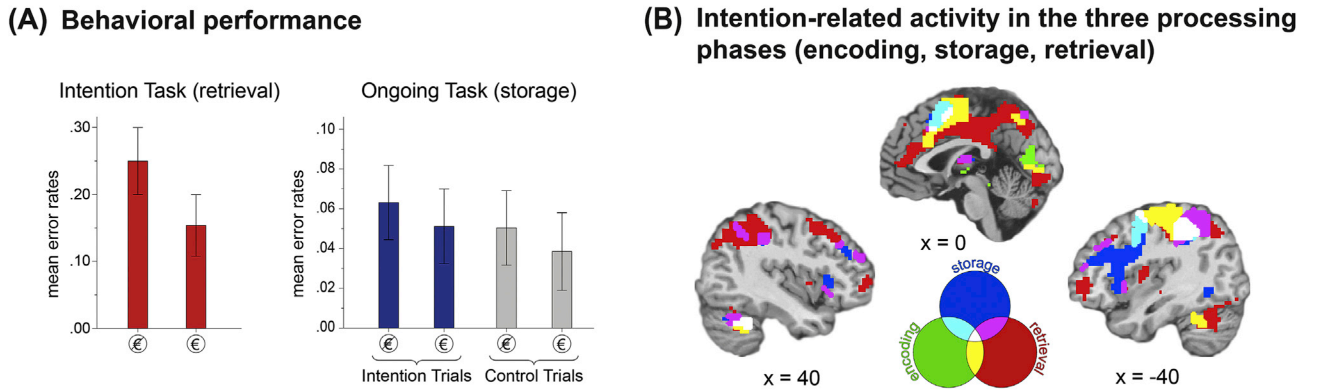

Fig. 2. (A) Behavioral results indicate improved intention retrieval and one-back performance in the rewarded trials. Error bars represent ± two standard errors. (B) Transient, phase-specific and overlapping activity changes in non-rewarded blocks defining the intention-related network ($p < .05$ FWE cluster-corrected).

$p < 0.001$, $\eta^2 = 0.83$, with longer responses in the encoding phase ($M = 531.13$ ms, $SE = 15.78$) than in the storage phase ($M = 467.77$ ms, $SE = 14.37$). Besides, the main effect of reward was significant, $F(1,21) = 17.12$, $p < 0.001$, $\eta^2 = 0.45$, with longer RTs in the rewarded condition ($M = 504.17$ ms, $SE = 14.45$), than in the non-rewarded condition ($M = 494.73$ ms, $SE = 15.09$). Additionally, the main effect of intention was also significant, $F(1,21) = 57.48$, $p < 0.001$, $\eta^2 = 0.73$, with longer responses in the intention trials ($M = 521.95$ ms, $SE = 15.98$), than in the control trials ($M = 476.95$ ms, $SE = 13.92$). None of the interactions reached the conventional level of statistical significance (all $p > .079$).

Functional imaging data

Intention-related effects

First, we tested on the whole brain level for regions associated with the processing of delayed intentions (intention trials) compared to pure ongoing task performance (control trials) in the non-rewarded context. This was done separately per processing phase (encoding, storage, retrieval). Note that the description of the results of the imaging data focuses on the frontal cortex. Detailed information concerning further brain regions can be found in the supplemental material (Supplementary Table S1).

As can been seen in Fig. 2B, the analysis of the main effect of intention widely replicated the findings of Gilbert (2011). Each processing phase was associated with specific but also with overlapping activity in medial and lateral frontal regions, in parietal cortex and in further posterior and subcortical regions (see Supplementary Table S1). Concerning PFC activity, the encoding of intentions was associated with medial PFC activity in regions of the pre-supplementary motor area (pre-SMA) and the anterior cingulate cortex (ACC). The storage of intentions during the delay interval also involved the pre-SMA. Most strongly, however, storage was associated with activity in the lateral PFC including BA 10 and more caudal PFC regions with more pronounced effects in the left PFC. Retrieval-related activity was also most prominent in medial and lateral frontoparietal regions. In contrast with the findings of Gilbert (2011), intention retrieval was also associated with two clusters in the bilateral lateral aPFC, which were located even more anteriorly than the storage-related activity.

The combined effect of all three task phases was used to create a binary mask image for testing brain-behavior correlations in frontal intention-related regions (see Supplementary Fig. S1).

Sustained reward effects and association with behavioral effects

Several frontoparietal and subcortical regions showed a sustained increase in activity in rewarded blocks compared to non-rewarded blocks (see Supplementary Table S2 and Fig. 3A). In the PFC, this included the ACC and pre-SMA but most strongly the bilateral lateral PFC, with large clusters ranging from BA10 and orbitofrontal cortex (OFC) along the inferior frontal sulcus to more posterior PFC regions. In addition, the anterior insula, bilaterally, was activated more in rewarded blocks compared to non-rewarded blocks as well as the right superior frontal sulcus. This sustained reward-related activity overlapped in large parts with the frontal intention-related network, but also included additional regions (see Supplementary Fig. S2).

To assess whether this sustained increase in activity was related to behavioral improvements we included the behavioral reward-related effects for the intention task (% improvement in ERs) as a covariate in the model to assess associations on a voxel-wise level. No cluster was present that showed a correlation with the behavioral reward effects in this analysis of sustained reward effects.

Transient reward effects and association with behavioral effects

The event-related analysis of reward effects revealed no general reward-related signal increases during encoding and retrieval. That is, in the whole group, the prospect of reward for correct intention retrieval did not specifically modulate these processing phases in the whole group. During the storage phase, reward led to increased activity in several regions, including bilateral aPFC, right dorsolateral PFC, ACC, superior parietal, occipital cortex and activity in the striatum (Fig. 3B, Supplementary Table S3). Note that this reward-related modulation of activity was not specific to intention trials but also present in control trials, as revealed by no significant interaction effect in any of these regions. Also, there was no correlation of this activity with the reward-related behavioral improvement in the intention task.

In contrast, an individual differences analysis revealed a circumscribed effect of reward in the right lateral aPFC specific to the encoding of intentions ($x = 33$, $y = 53$, $z = 22$, $t_{max} = 5.86$, $k = 42$), that is, individuals with stronger behavioral reward effects also showed stronger reward-related activity increases in this region while encoding the intention (see Fig. 3C). Post-hoc analyses revealed, that the activity cluster overlapped completely with the retrieval-related activity in the right lateral PFC present in the non-rewarded condition (cf. Fig. 2B). No such correlation was present for control trials within the frontal intention-related mask. The encoding-related effect in this cluster is further illustrated in Fig. 3D - showing that the correlation with the behavioral reward effects is present in intention trials but not in the control trials where only the one-back task was performed.

Thus, transient activity changes on the group level during storage were related to the prospect of reward in general but not specific to intention processing. In contrast, individual differences analyses revealed that the additional transient recruitment of the right lateral aPFC during encoding was related to the improved retrieval of delayed intentions associated with reward.

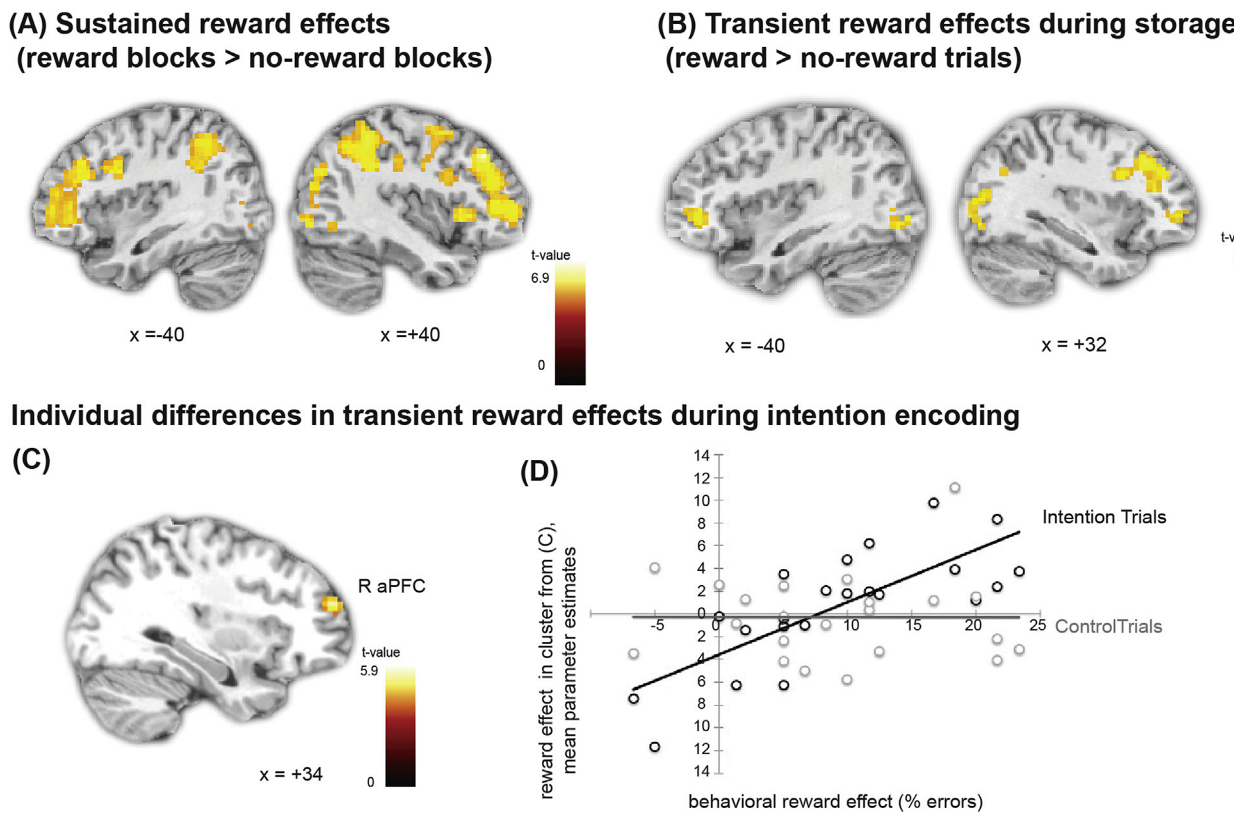

Fig. 3. Sustained and transient reward effects and association with behavioral reward effects. (A) Sustained effects. Increased activity in rewarded blocks compared to non-rewarded blocks ($p < .05$, FWE-cluster corrected). (B) Transient reward effects during storage phase. (C) Individual differences in transient reward effects during intention encoding. Correlation of encoding-related activity in rewarded intention trials compared to non-rewarded intention trials (within frontal intention-related network) with behavioral reward effects in the intention task ($p < .05$, FWE-SVC-cluster corrected). (D) Illustration of brain-behavior correlation from mean parameter estimates of all voxels in cluster in (C) and qualitative comparison to correlation in control trials.

Discussion

The aim of the present study was to investigate whether increased motivation modulates the processing of intentions in brain regions associated with different phases of intention processing, whether this is associated with sustained or transient neurocognitive mechanisms and whether this is directly related to behavioral reward-related improvements.

The behavioral data indicate a striking improvement in intention-retrieval performance based on the prospect of a monetary reward. This confirms that motivation can indeed modulate the ability to realize delayed intentions as previously shown in several behavioral prospective memory studies (Kliegel, 2004; Meacham and Singer, 1977; Walter and Meier, 2014).

Importantly, the imaging data provide evidence for the mechanisms underlying this improvement. In the whole group, the prospect of a monetary reward for correct intention retrieval evoked sustained but also transient activity changes. The sustained effects were rather widespread, including increased reward-related activity in bilateral fronto-parietal regions, including the lateral aPFC, as well as subcortical regions. Transient reward effects in the whole group were restricted to the storage phase within subregions of this network, focused in frontoparietal and occipital regions. In addition, individual behavioral reward effects in the intention task were directly associated with stronger transient encoding signals in the right lateral aPFC. Individuals who showed a selective enhancement in brain activity in this region during the encoding of reward-associated intentions also showed improved intention retrieval performance compared to the non-rewarded condition. Note, that this effect was only present when applying a small volume correction for multiple comparisons, focusing on frontal intention-related regions. Thus, both sustained and transient increases in lateral aPFC seem to support the retrieval of rewarded intentions.

Motivation-cognition interactions for processing delayed intentions

The understanding of the mechanisms underlying the improvement in the retrieval of delayed intentions associated with reward prospect is interesting from various perspectives.

On the one hand, models discussing importance effects on prospective memory (Penningroth and Scott, 2013) suggest various changes in intention processing related to the importance of an intention. A fundamental effect might be an increase in goal relatedness, that is, importance adds greater personal relevance to the intention, in our case the potential to earn money for good performance. This more general effect is assumed to affect intention processing on several levels - such as increasing accessibility in memory (Jeong and Cranney, 2009) or enhancing monitoring processes for cue retrieval (Kliegel et al., 2004). Our data support the view that reward affects intention processing on several levels. On a more global level, this is reflected in the sustained reward-related signal increases in fronto-parietal regions. These regions were previously shown to be related to motivated cognition (Jimura et al., 2010) and cognitive control (Braver et al., 2003) and might reflect a rather unspecific reward-related increase in task engagement (Dosenbach et al., 2008), which might then be beneficial for several component processes of intention processing. Note, however, that the sustained increase in activity was not directly associated with individual improvements in intention retrieval performance. In contrast, the phase-specific transient changes during the encoding of intention-related stimuli in the right aPFC were correlated with behavioral improvements and might reflect the more fine-grained adjustments required for successful intention encoding. Thus, our imaging data further specify how these different mechanisms supporting motivated intention processing might be implemented in the brain and where individual differences might arise from.

On the other hand, the current findings are further interesting from the perspective of resource theories, which commonly assume that interference between two tasks arises when these require common processing resources (McDaniel and Einstein, 2000; Navon and Gopher, 1979; Wickens, 1991). The data of the non-rewarded condition clearly show that such interference effects were present in our paradigm - participants showed decreased performance in the n-back task when an additional intention was present compared to a situation where the n-back task was performed alone. Note, however, that the improved performance in the rewarded intention task did not come at costs for the ongoing n-back task, which might have been expected from a resource perspective. In contrast, we also found decreased ERs in the ongoing task in rewarded blocks and this was not only the case during sequences when the rewarded intention was present but also during control trials. Thus, in contrast to the detrimental effects on ongoing task performance for importance manipulations concerning the delayed intention task shown in a previous study by Kliegel (2004), we found a generalized effect of motivation on both the delayed intention task and the ongoing task performance. This might suggest that, despite the presence of interference between tasks, participants did not work at the limits of their resources in the non-rewarded condition. This finding is in line with different psychological and neurocognitive models. For example, classical models in motivation psychology describe motivation as including a globally energizing factor that varies independently of control demands and behavioral direction (Duffy, 1962; Hull, 1943).

In addition, the Dual Mechanisms of Control model (Braver, 2012) postulates the existence of a proactive and a reactive cognitive control mode for dealing with interference and conflict. One central prediction of that model is that reward-related situations will lead to a shift towards a proactive control strategy, which is characterized by sustained maintenance and implementation of control in the lateral PFC (Braver, 2012; Locke and Braver, 2008). Crucially, this should also lead to enhanced performance in trials, which are presented in a rewarded context without actually being rewarded, as it was the case for our ongoing task trials. This has been shown previously on a behavioral and brain level (Jimura et al., 2010) with activity dynamics in the right lateral PFC supporting both, the sustained and the transient processing changes related to motivational incentives. Thus, while we find phase-specific transient activity changes during intention encoding in the right lateral aPFC, the sustained bilateral PFC activity throughout the block seems to have its beneficial effect on the non-rewarded ongoing task as well. Whether or not a different pattern for the ongoing task might be obtained in a situation with a more demanding ongoing task, which requires more resources, would be interesting to test in future studies.

Note that the phase-specific transient activity changes during intention encoding in the right lateral aPFC was only present in the individual differences analysis while no encoding-related effect was present when averaged across all participants. Thus, individuals showing a higher intention retrieval performance under reward seemed to recruit the right lateral aPFC during encoding in addition to other encoding-related regions. Interestingly, this region overlapped with the retrieval network identified in the whole group. This supports previous findings indicating beneficial effects for retrieval depending on the overlap between encoding- and retrieval-related activity (Gilbert et al., 2012). Our findings further support the benefits of individual differences analyses in revealing underlying brain mechanisms that might be obscured in group analyses (Yarkoni and Braver, 2010).

Strategic monitoring vs. spontaneous retrieval in prospective memory

Recent discussions suggest that the degree to which motivational incentives for a prospective intention task have beneficial vs. detrimental effects on an ongoing task might depend on the type of motivational incentive (Walter and Meier, 2014). Specifically, Walter and Meier (2014) suggest that certain motivational incentives might lead to an increase in controlled monitoring for intention-related information whereas others might build grounds for a spontaneous retrieval strategy based on strong encoding. While the beneficial behavioral effects in the ongoing task of the present study seem to indicate a reduction in monitoring and may suggest a switch to a spontaneous retrieval strategy, the sustained reward-related increase in activity in lateral aPFC regions, which has been related to monitoring processes (Beck et al., 2014; Gilbert, 2011; McDaniel et al., 2013) contradicts such an interpretation. Note that McDaniel et al. (2013) found sustained activation of lateral aPFC specifically for non-focal prospective memory tasks (i.e., the intention task does not focus on the same stimuli features as the ongoing task). This was different in the present study, where the relevant stimulus feature (word identity) was the same for the intention task and the ongoing task. Despite the focality of the task, we observed behavioral interference effects in the ongoing task and also sustained activation during the storage phase in the non-rewarded condition. So, there might be monitoring involved - however this is not specifically modulated via rewards during the storage phase but rather in an unspecific block-wise manner.

On the other hand, transient neural effects in right lateral aPFC were specific to the intention-encoding phase, suggesting an increased encoding strength in the rewarded condition, which would be in line with the involvement of a spontaneous retrieval strategy.

Accordingly, the present data suggest that the two strategies might not necessarily be mutually exclusive - motivational incentives seem to provide a basis for strengthening the encoding of intentions, which is a prerequisite for spontaneous retrieval. However, participants seem to invest additional effort in a sustained manner, maybe to assure that they do not miss the intention retrieval cue and the according reward. This, however, is not specific to the storage phase but reflects a rather unspecific, global effect of motivation on cognition on the block level. As with the present paradigm, we cannot disentangle reward-related activity associated with the intention task compared to activity associated with the ongoing one-back task, conclusions about reduced vs. increased intention-related monitoring must be made with caution, especially as we also found improved one-back performance in rewarded blocks. Note that this generalized motivation effect might also be related to the performance criterion for the ongoing task (80% correct) that was included to prevent participants from engaging only in the intention task in order to achieve the maximal bonus. Maybe the improvement in the intention task might have been even stronger without this criterion, together with performance decrements in the ongoing task. This might be interesting to test empirically.

Furthermore, the event-related nature of our design with only minimal perceptual differences between intention trials and control trials might have further supported the emergence of sustained effects - perhaps participants were not sure whether the current trial was a control trial or an intention trial and therefore tried to monitor in control trials as well. However, the transient effects specific to intention encoding still suggest that individuals who were able to selectively enhance their encoding under reward were also more successful in retrieving their intentions.

Modulation of stimulus encoding in different memory domains

Reward-related effects on the encoding of stimuli into memory have been shown previously in the domains of working memory and long-term memory. For working memory, increased activity in stimulus-specific association cortices and lateral PFC was shown for stimuli associated with high reward (Jimura et al., 2010; Krawczyk et al., 2007). In the domain of long-term memory, it has been shown that activity in the hippocampus and the midbrain during the encoding of stimuli is higher for those rewarded stimuli that are actually recognized 24 h (Adcock et al., 2006) to three weeks later (Wittmann et al., 2005). Notably, functional connectivity analyses revealed a potential neuromodulatory effect on memory encoding via cortical regions (Cohen et al., 2014; Murty and Adcock, 2014; Shigemune et al., 2017). For example, Murty and Adcock (2014) showed that encoding-related activity associated with reward in the hippocampus was preceded by interactions of dopaminergic midbrain regions with prefrontal and visual regions, providing a potential indirect neuromodulatory mechanism for hippocampus-dependent memory associated with reward.

Our findings are in line with these findings from other memory domains, showing that the regions actually involved in the respective memory processes are recruited to a higher degree when the relevant stimulus is related to reward. For the encoding of intentions, this is the lateral aPFC, which plays a crucial role in storing and retrieving delayed intentions as shown in the respective contrasts in the non-rewarded blocks. Recruiting these regions during encoding already seems to be crucial for benefiting from the reward on a behavioral level. This finding further adds to existing data on dissociable networks for working memory and prospective memory (Reynolds et al., 2009; West et al., 2006). Despite obvious similarities between the applied working-memory task and the intention task, which both involved the temporary storage of items, reward effects for the intention task were located in intention-related regions in aPFC rather than in working-memory regions. This specificity might be related to those aspects that are specific to the processing of delayed intentions, for example that an interfering ongoing task has to be processed or that retrieval has to be self-initiated. Alternatively, one might argue that this aPFC effect is related to other processes engaged in processing the encoding cue, such as salience processing or switching from the ongoing task to the intention task (Bisiacchi et al., 2009). Increased reaction times for the ongoing task responses during encoding compared to storage might reflect these additional demands. While we are confident that we ruled out salience processing by applying a comparably salient color cue in the control condition, we cannot separate processes related to switching from the ongoing task to intention encoding, the actual intention encoding process and dealing with these processes in an overlapping manner. However, as the brain-behavior correlation was specific to the encoding phase while switching processes would be even more relevant during the retrieval phase, where a different response needs to be initiated, we suggest that reward affects intention encoding per se albeit not necessarily in an exclusive manner. This temporal separation of processes within one trial could be an endeavor for future electroencephalographic studies with higher temporal resolution (West et al., 2006).

Finally, whether or not effects in the aPFC reflect a direct effect of reward-related dopaminergic modulation or rather an indirect modulatory effect via more caudal and/or medial prefrontal regions needs to be tested in larger samples in the future. A contribution of cognitive control mechanisms to enhanced encoding of rewarded intentions seems plausible from models focusing on motivation-cognition interactions in prospective memory (Penningroth and Scott, 2007).

Involvement of lateral aPFC in retrieval

Brain activation patterns associated with the intention task in each phase widely replicated the findings by Gilbert (2011). However, there were also some deviations from the pattern, the most relevant being an involvement of the very anterior part of the lateral aPFC in the retrieval of intentions, which was not present in the Gilbert study. While these bilateral clusters were more circumscribed compared to the storage-related cluster, which also covered more posterior parts of the lateral PFC, their presence seems to contradict the conclusion that solely the storage of intentions is related to activity in the lateral aPFC. We can only speculate about the reason for this different finding. One methodological difference might be MRI field strength (3 T in the present study vs. 1.5 T in Gilbert's study) which might have changed the sensitivity for some effects. In addition, we used a tilting of slice positioning and field maps to optimize anterior PFC coverage, which might have further increased the sensitivity to these effects in our study. In addition, there were some special characteristics to the stimulus materials, related to the applied decoding approach in the Gilbert study. In particular, different stimuli (words and pictures) were mixed in that study while we exclusively used word stimuli. This might have changed the strength of the retrieval signal and/or the retrieval strategy applied by the participants. As outlined above, this retrieval-related signal was spatially close to the transient reward-related effects during encoding, although somewhat more medial. This suggests that reward might have modulated the similarity in activity between encoding and retrieval, which has previously been shown to be beneficial for intention retrieval (Gilbert et al., 2012).

Conclusion

The current study provides insights into the neural mechanisms subserving the motivational modulation of processing delayed intentions. Our findings suggest that the ability to mobilize and implement control processes during the encoding and storage of intentions is crucial for motivational effects on successful intention retrieval, in addition to a general increase in processing effort in a frontoparietal network. Moreover, they indicate that bilateral intention-related regions in the lateral aPFC are relevant for these motivation-cognition interactions.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.neuroimage.2018.01.083.

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